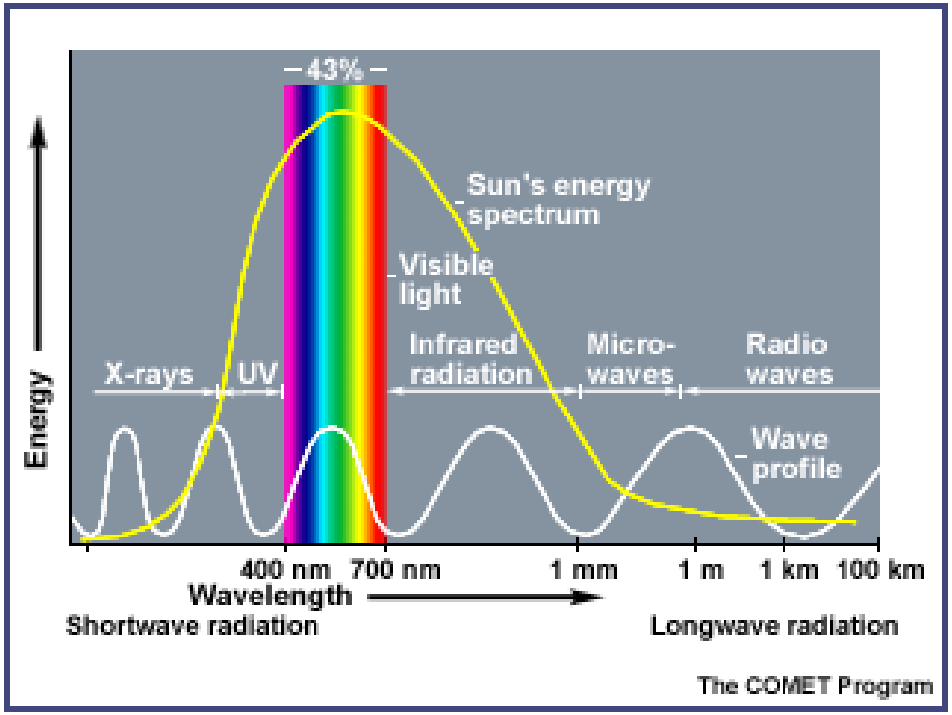
Solar Photovoltaics

What is Solar Energy?

- Originates with the thermonuclear fusion reactions occurring in the sun.
- Represents the entire electromagnetic radiation (visible light, infrared, ultraviolet, x-rays, and radio waves).
- UV: 7.6% (150-380 nm)
- Visible:48.4% (380-720nm)
- Infrared: 43% (720-4000nm)



Solar Radiation

- Solar power generated using PV systems is intermittent and variable

- Power produced by a PV system depends mainly on absorbed solar radiation–Global Solar Radiation (GSR) forecasting tool is essential

- Knowledge of GSR helps in anticipating problems related to the nature of PV power and helps in pre-planning for minimizing influence of fluctuations

Ventura and Tina, 2015

$$P = P_{STC} \frac{G_{POA}}{G_{STC}} \left[1 - \alpha_P (T_{PV} - T_{STC})\right]$$

$$T_{PV} = T_{amb} + (NOCT - 20) \frac{G_i}{800}$$

STC-Cell temperature of 25°C, irradiance of 1000 W/ sq.m.

NOCT- Temp. reached by a cell under following operating conditions:

Irradiance on cell surface = 800 W/m^2

Air Temperature = 20°C

Wind Velocity = 1 m/s

Global Solar Radiation (GSR)

- Direct radiation/beam Radiation
  - Without absorption or scattering
- Diffuse Radiation
  - Scattering interaction with earth's atmosphere
- Albedo Radiation
  - Reflected Radiation

Estimation of Solar Radiation

- Various methods to find out/estimate Solar Radiation
  - Direct measurement using Pyranometer (expensive equipment and data acquisition set up. Installed at only very selected locations, mostly metropolitans)
  - Satellite based methods
    - Expensive, lack of historical data, forecasting errors
  - Empirical Models
    - Use correlation between solar radiation and geographical and meteorological parameters.
  - Stochastic algorithm models
  - AI techniques
    - NN,ANFIS etc.

Solar Radiation

- Solar Radiation Energy (Irradiation)
  - Amount of energy received by a surface over a period of time (W-Hr/sq.m.) per hour/day/month.
- Solar Radiation Power (Irradiance)
  - Rate of Energy received by surface per unit area (W/sq.m.)

Irradiance

- Rate of Energy received by a surface per unit area. (W/Sq.m.)

Irradiance/Insolation

- Amount of energy received by a surface over a given period of time. (W/Sq.m. per hour/day/month etc.)

Solar Constant

- Amount of Energy that reaches just outside earth's atmosphere. (1367 W/Sq.m)
- Measured at the surface perpendicular to the sunrays at the average sun-earth distance on the top of the atmosphere.
- Sun-earth distance varies throughout the year.

$$S_t = S\left[1 + 0.033\cos\left(\frac{360n}{365}\right)\right]$$

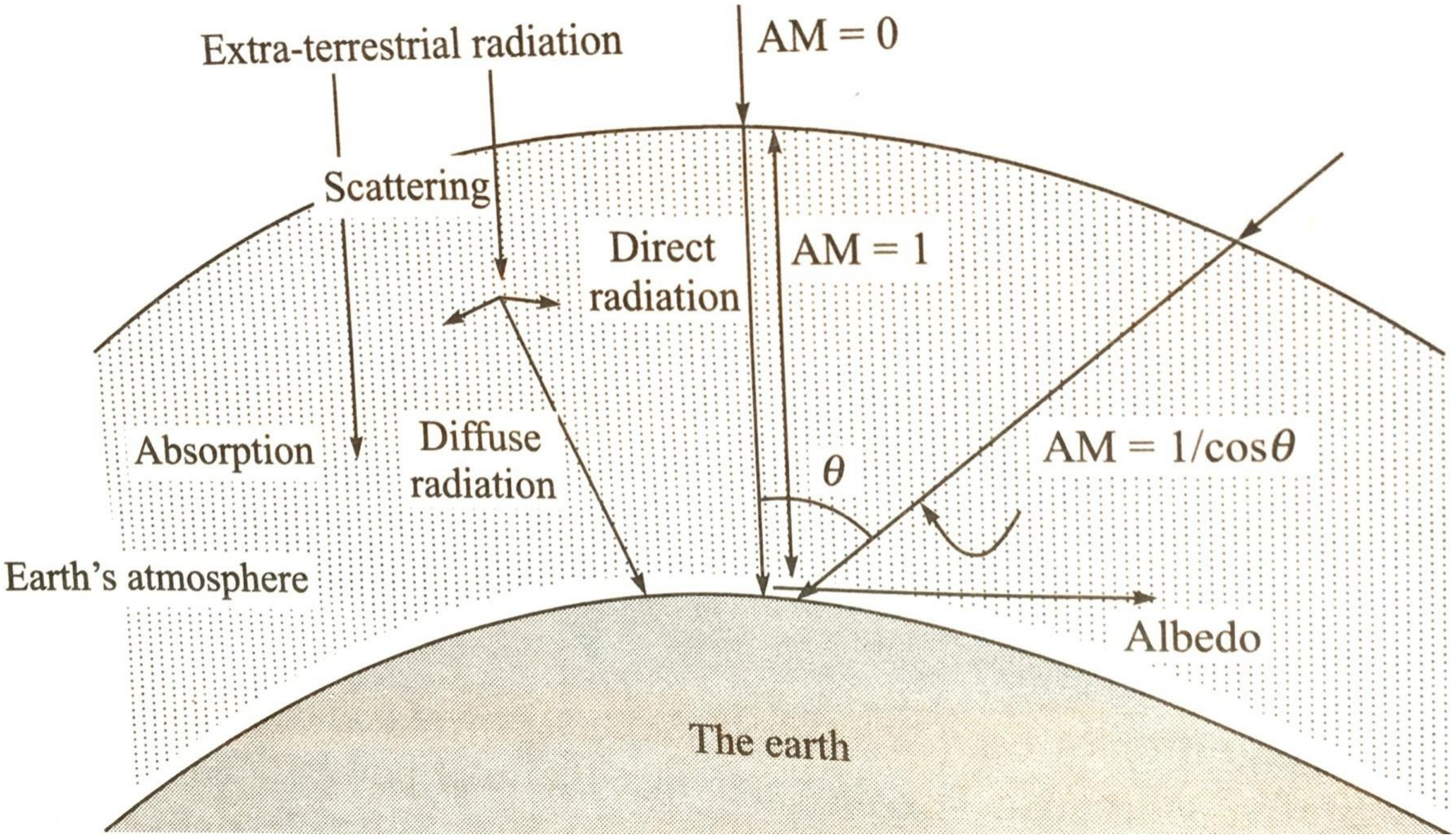
Global Solar Radiation (GSR)

- Direct radiation/beam Radiation
  - Without absorption or scattering
- Diffuse Radiation
  - Scattering interaction with earth's atmosphere (15-20% of Direct)
- Albedo Radiation
  - Reflected Radiation

Air Mass

- AM: distance travelled by the sun rays in the earth's atmosphere. AM= $1/\cos\theta$
- AM0: radiation spectrum just outside earth's atmosphere (1376 W/sq.m.)
- AM1: (1105 W/sq.m.)
- AM1.5: (1000 W/sq.m.)
- AM2: (894 W/sq.m.)

AM



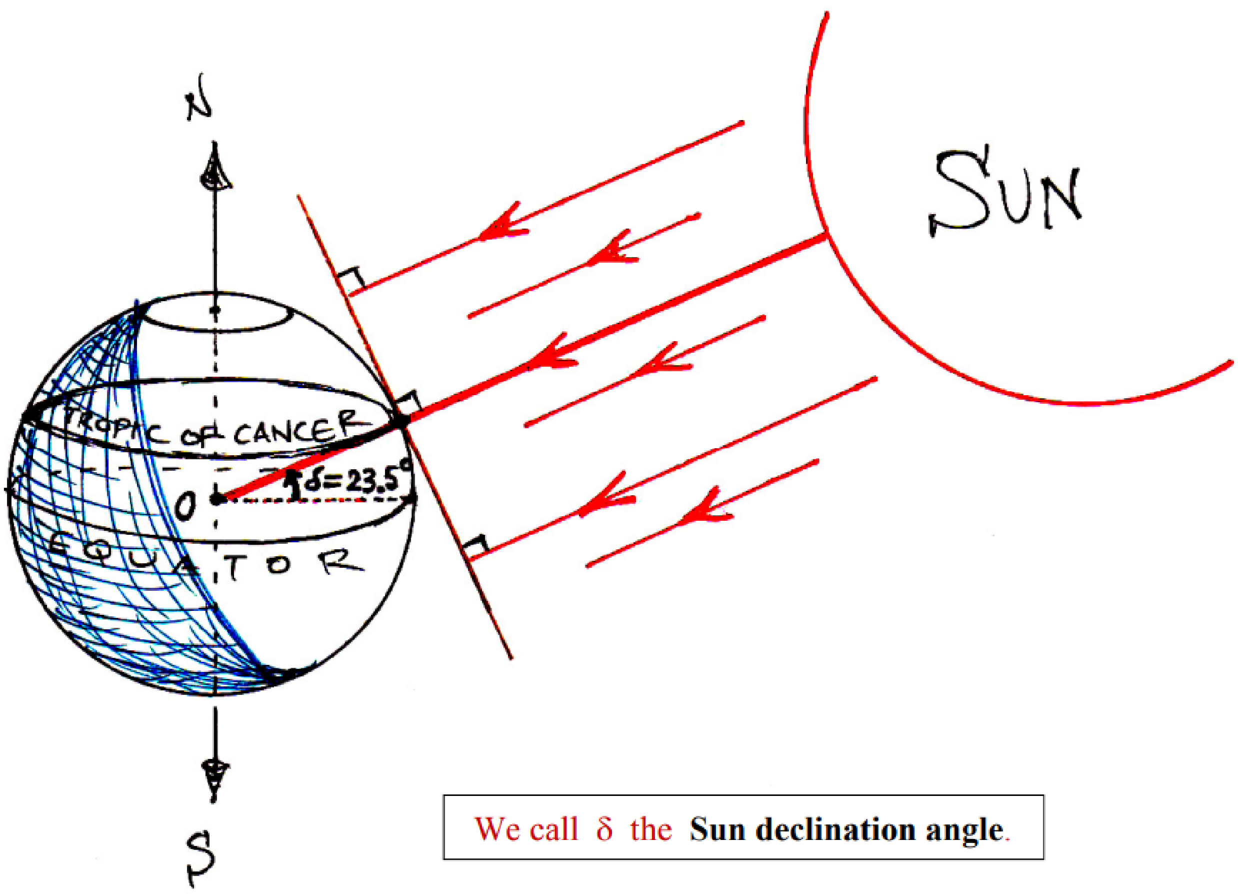
Extra-terrestrial radiation
AM = 0
Scattering
Direct radiation
AM = 1
Absorption
Diffuse radiation
θ
AM = 1/cosθ
Earth's atmosphere
Albedo
The earth

Declination Angle

- Angle between: Lines joining the centre of the earth to the centre of the sun with its projection on the equatorial plane.
- Declination angle varies due to inclination of the earth's polar axis and its revolution around the sun.

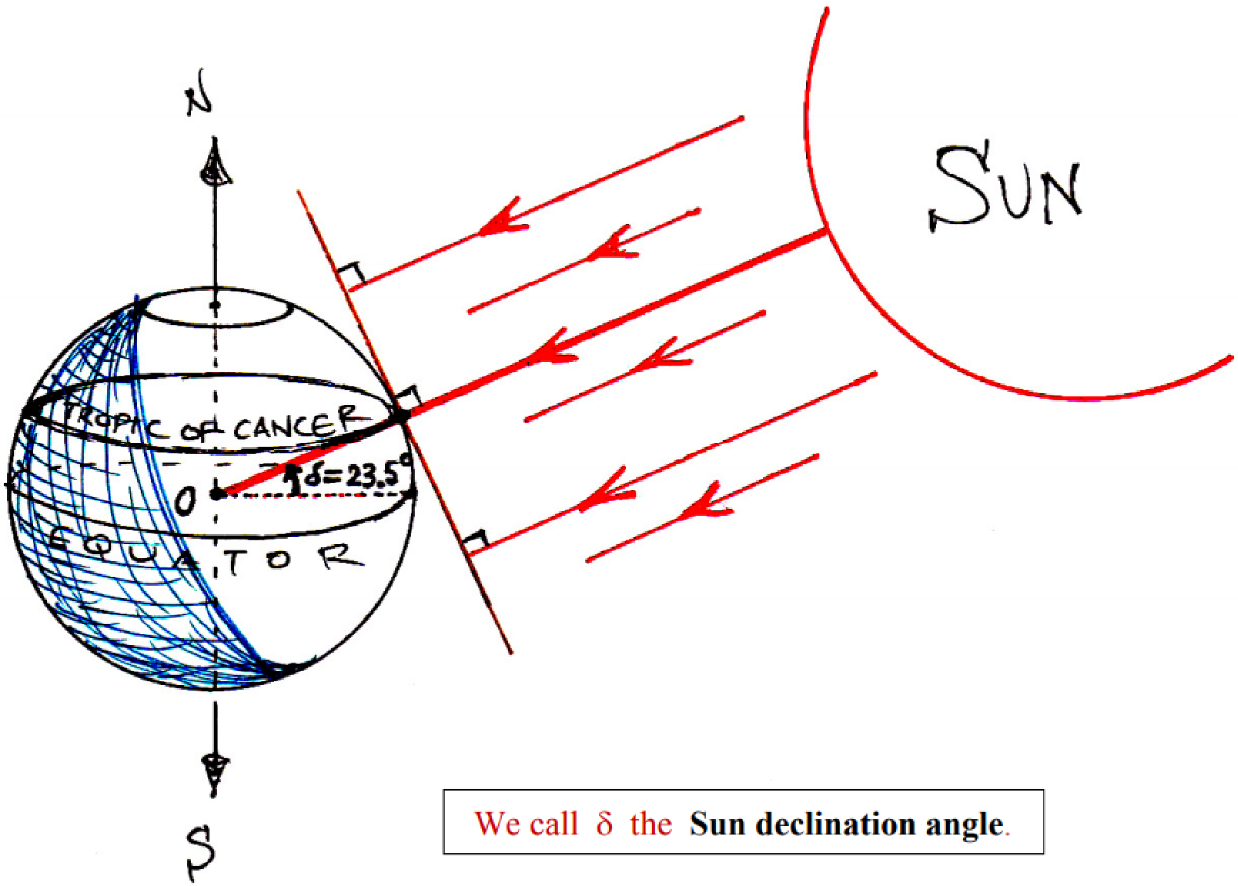
$$\delta = 23.34 \sin\left[\frac{360}{365}(284 + n)\right]$$

Declination Angle (June 21)



We call δ the **Sun declination angle**.

Declination Angle (Dec. 22)



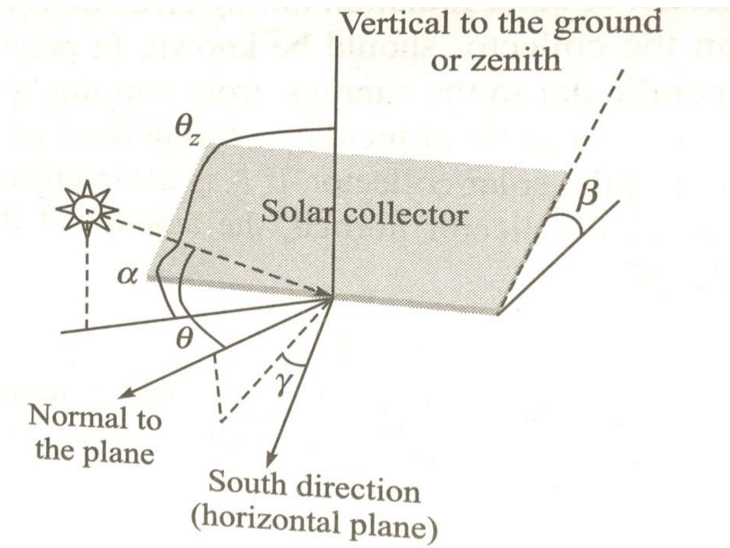
We call δ the **Sun declination angle**.

Angle of Sunrays

- Angle of Incidence: Angle Between the direct Sunrays and normal to the collector
- Latitude (Ø) (Angle between the lines joining the earth's centre to the location with its projection on the equatorial plane. (Northen Hemisphere: positive angle)
- Time of the Day (Omega): Represented by hour angle. 1 hour = 15 degrees
- Inclination of Surface (Beta): Angle made by the collector surface with the horizon (0-180)

Orientation of Collector in Horizontal Plane

- Angle Between the lines due south and the projection of normal to the collector on the horizontal plane **(γ)**



$$\cos\theta = \sin\phi(\sin\delta\cos\beta + \cos\delta\cos\gamma\cos\omega\sin\beta) + \cos\phi(\cos\delta\cos\omega\cos\beta - \sin\delta\cos\gamma\sin\beta) + \cos\delta\sin\gamma\sin\omega\sin\beta$$

Relationship between various Angles

- Situation 1: Collector flat on the ground
  - Inclination of Surface (β=0)
- Situation 2: Collector Surface is facing south
  - Surface Azimuth Angle (γ)=0

Exercise 1

- What will be the angle of Incidence in Mumbai at Noon on November 01 on Horizontal Plane!

Local Apparent Time (LAT)

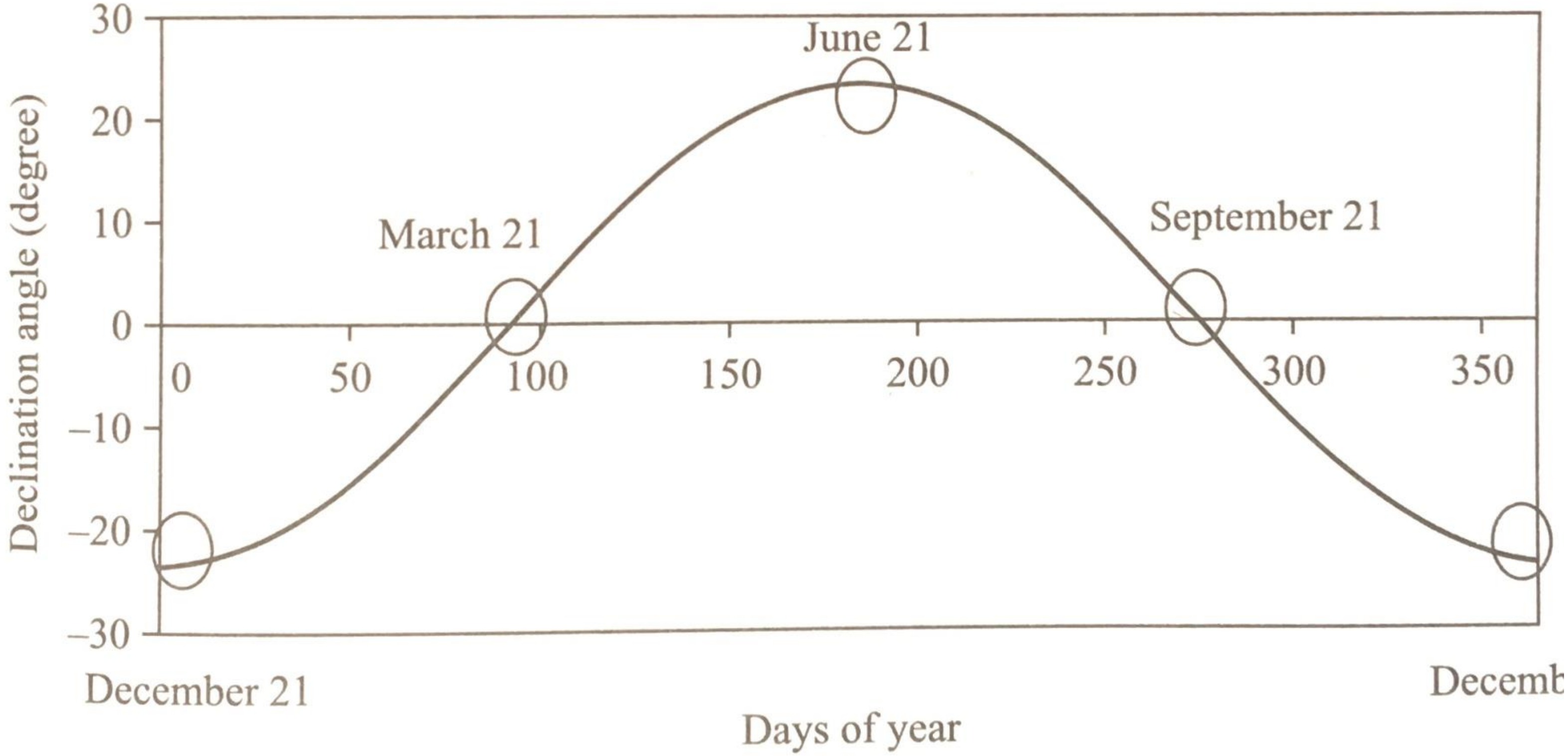
- Time at which the earth becomes overhead at a given location.
- Standard time for a country is based on Noon at a given longitude.
- For a location on a different longitude in the same country, the noon time needs to be corrected.
- 1Hour = 15 degrees of longitude difference.
- 1 Degree longitude= 4 min difference in time.

LAT

- Earth's rotation speed and orbit are subject to some fluctuations. The amount of time to be corrected varies from month to month

$$\text{LAT} = T_{\text{local}} + \frac{1}{15}(\text{Long}_{\text{st}} - \text{Long}_{\text{local}}) + \frac{1}{60}\ \text{Equation of time}$$

Declination Angle variation



Exercise 2

- Calculate the declination angle, LAT and hour angle for the collector located in Bombay (19.12 N, 72.51 E) which is tilted at an angle of 30 degrees and is pointing due South on October 1.

Solar Radiation Estimation

- Monthly Average **Daily Global radiation** on Horizontal Surfaces

$$\frac{H_{ga}}{H_{0a}} = a + b\left(\frac{S_a}{S_{\max a}}\right)$$

- Hoa : Monthly average **extra terrestrial solar radiation** on horizontal surface
- Sa: monthly average daily sunshine hours
- Smaxa
- Ho=Hoa (for a given day)

Solar Radiation Estimation

$$H_0 = S_t \int \cos\theta \, dt$$

$$= S\left(1 + 0.033\cos\frac{360n}{365}\right)\int_{\text{sunrise}}^{\text{sunset}} (\sin\phi\ \sin\delta + \cos\phi\ \cos\delta\ \cos\omega)\ dt$$

Solar Radiation Estimation

$$dt = \frac{180}{15\pi} d\omega$$

$$H_0 = \frac{12}{\pi} S \left(1 + 0.033 \cos \frac{360n}{365}\right) \int_{-\omega_s}^{\omega_s} (\sin\phi \, \sin\delta + \cos\phi \, \cos\delta \, \cos\omega) d\omega$$

$$H_0 = \frac{24}{\pi} S \left(1 + 0.033 \cos \frac{360n}{365}\right) (\omega_s \sin\phi \, \sin\delta + \cos\phi \, \cos\delta \, \sin\omega_s)$$

Exercise 3

- Estimate the monthly average daily global radiation on the horizontal surface at Nagpur (21.06N, 79.03E) during the month of March if the average sunshine hours per day is 9.2. Assume ($a=0.27$ and $b=0.50$)